

The Discrete RVs

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MATH 241

Week 6, Lecture #1 — Ch. 3.3-3.5

Reminders!

- Homework #4 due on Sunday at 11:59pm
 - ▶ Textbook problems plus a mini-investigation of the Poisson distribution
- Checkpoint #3 at beginning of this Thursday's class
 - ▶ More activity-like than quiz-like; in group settings
- New office hours starting this week:
 - ▶ Monday from 1-2pm
 - ▶ Tuesday from 3-4pm
 - ▶ Friday from 10-11:30am

Common Distributions

A random variable X may involve similar characteristics and traits, such as...

- the use of repeated trials
- the density of observations
- the components of the situation

As such, we have common distributions (both of the discrete and continuous type). The benefits?

- Same PMFs/PDFs, CDFs, expectations, variance, etc.
- Flexibility in result dependent on values of population parameters
 - ▶ Will be seen as what appears to be a conditional probability
 - ★ e.g. $P(X = 3|n = 10, p = 0.25)$

Common Distributions

Certain situations have fortunately been found as valid probability distributions. For the discrete RVs, we have the following:

- **Bernoulli**: a single binary outcome, with some parameter p of an outcome occurring
- **Binomial**: n independent, binary trials, with some parameter p of an outcome occurring
 - ▶ Multinomial: extension to n independent trials of multiple outcomes with probability $p_1, p_2, \dots, p_n$ occurring
- **Geometric**: first success from binary, independent trials with some parameter p of an outcome occurring
- **Hypergeometric**: k successes from n draws without replacement with K possible successes and N outcomes
- **Discrete uniform**: uniform probability across n values of support, where probability is equal to $1/n$

The Bernoulli

- We might care about a **single** trial of binary outcomes
 - ▶ A **Bernoulli experiment**
 - ▶ e.g. success/failure
 - ▶ e.g. yes/no

The Bernoulli

Start with some binary trial, with a success and failure as outcomes. There is some probability of success p and probability of failure $1 - p$.

Then, some random variable X has a Bernoulli distribution with parameter p , where $0 < p < 1$, if:

$$\begin{aligned} P(X = x) &= \begin{cases} 1 - p & \text{for } x = 0 \\ p & \text{for } x = 1 \end{cases} \\ &= p^x(1 - p)^{1-x} \quad \text{for } x \in \{0, 1\} \end{aligned}$$

Denote this as:

$$X \sim \text{Bern}(p)$$

The Bernoulli

Expectation (i.e. average) and variance:

$$E(X) = 0 \times (1 - p) + 1 \times p = p$$

$$\text{Var}(X) = E(X^2) - E^2(X) = p - p^2 = p(1 - p)$$

We will talk about deriving the expectation and variance later. However, these will be other traits of a Bernoulli distribution.

Question: Suppose we run a Bernoulli experiment n number of times, independent from each other. What happens?

The Binomial

We tend to care more about repeating this binary trial many, many times independently. We add a new parameter to this new setup: n , the number of independent binary trials completed.

We now care about x successes* from n trials.

This describes a **binomial** distribution, where $X \sim \text{Bin}(n, p)$ and its PMF is:

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

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Why? Consider this:

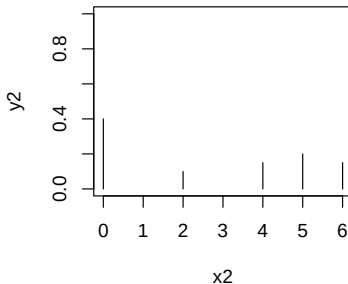
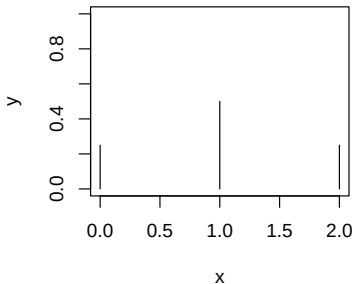
- At $n = 4$ trials: For two successes, we will need to:
 - ▶ Find the probability of two successes: p^2
 - ▶ Find the probability of two failures: $(1 - p)^2$
 - ▶ Find the number of combinations: $\binom{4}{2}$

The Binomial — PMF Plots

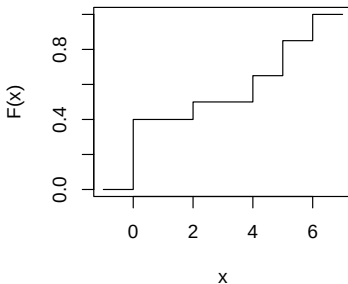
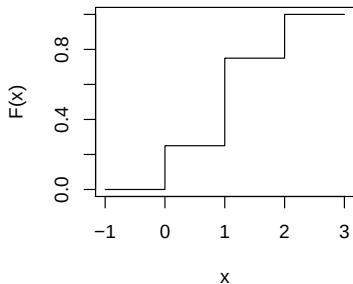
Expectation (i.e. average) and variance:

$$E(X) = np$$

$$\text{Var}(X) = np(1 - p)$$

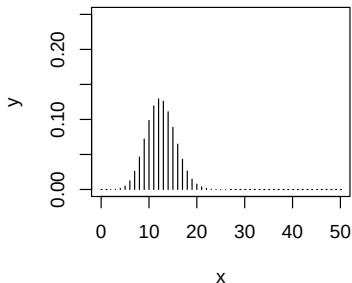


The Binomial — PMF Plots

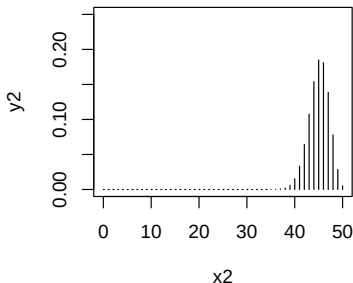


The Binomial — PMF Plots

$X \sim \text{Bin}(n = 50, p = 0.25)$

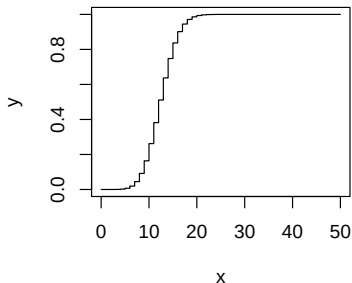


$X \sim \text{Bin}(n = 50, p = 0.9)$

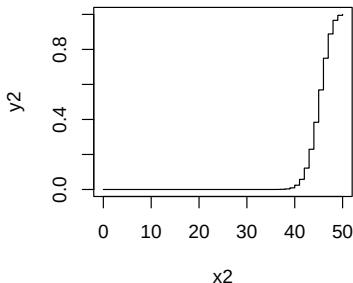


The Binomial — CDF Plots

$X \sim \text{Bin}(n = 50, p = 0.25)$



$X \sim \text{Bin}(n = 50, p = 0.9)$



Example: 4 Coin Flips

Say we flip a coin 4 times and let X represent the number of heads we get. Find the PMF and CDF for any value of x , then find $P(X = 2)$ and $P(X \leq 2)$.

The PMF and CDF are as follows, setting $n = 4$ and $p = 0.5$:

$$P(X = x) = \binom{4}{x} 0.5^x 0.5^{4-x}$$

$$P(X \leq x) = \sum_{a \leq x} \binom{4}{a} 0.5^a 0.5^{4-a}$$

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$P(X = 2)$ and $P(X \leq 2)$ are:

$$\begin{aligned}P(X = 2) &= \binom{4}{2} 0.5^2 0.5^2 \\&= 0.375\end{aligned}$$

$$\begin{aligned}P(X \leq 2) &= \binom{4}{0} 0.5^0 0.5^4 + \binom{4}{1} 0.5^1 0.5^3 + \binom{4}{2} 0.5^2 0.5^2 \\&= 0.6875\end{aligned}$$

Activity: Weighted Coin Flips of 10

Say we flip a weighted coin 10 times and let X represent the number of heads we get. Here, the coin is weighted such that there is a 70% chance to get a heads. Find the PMF and CDF for any value of x , then find $P(X = 3)$ and $P(X \leq 3)$.

Other Common Discrete Distributions

- **Multinomial:** greater than 2 outcomes (i.e. k possible outcomes)

- ▶ $\sum_{i=1}^k p_i = 1$

$$P(X = (x_1, x_2, \dots, x_k)) = \frac{n!}{x_1! x_2! \dots x_k!} p_1^{x_1} p_2^{x_2} \dots p_k^{x_k}$$

- **Hypergeometric:** sampling without replacement with equal chances of removal

- ▶ k successes from K possible successes
- ▶ n draws from N possible draws

$$P(X = x) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

- ▶ Wait a minute?! I've used this...

Other Common Discrete Distributions

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$$P(X = x) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}$$

- ▶ Wait a minute?! I've used this...
- ▶ Picking k freshmen from K freshmen and $n - k$ remaining spots from $N - K$ others

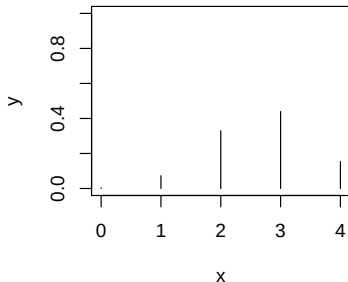
Other Common Discrete Distributions

- **Geometric:** first success for repeated, independent binary trials

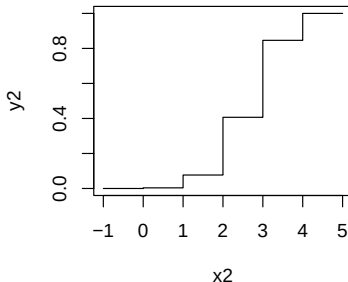
$$P(X = x) = p(1 - p)^{x-1}$$

PMF/CDF of Hypergeometric

$X \sim \text{Hyper}(m = 10, n = 5, k = 4)$

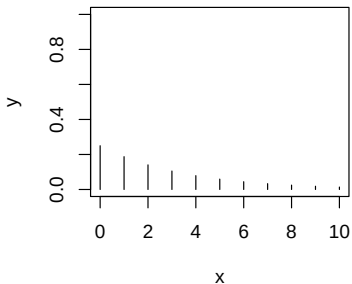


$X \sim \text{Hyper}(m = 10, n = 5, k = 4)$

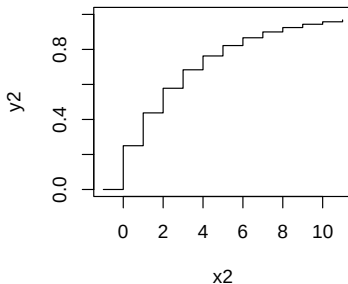


PMF/CDF of Geometric

$X \sim \text{Geom}(p = 0.25)$



$X \sim \text{Geom}(p = 0.25)$



Examples

- ① A manager at an electronics manufacturing plant receives a shipment of 20 microchips. Unknown to the manager, 5 of the chips are defective and 15 are non-defective. For the quality check, the manager randomly selects 4 chips from the shipment without replacement. Find:
- ▶ the probability that exactly 2 of the selected chips are defective.
 - ▶ the probability that less than 2 of the selected chips are defective.

Examples

- ② Brandon decides to enter the lottery repeatedly for hiking Angels Landing in Zion National Park, hoping he'll eventually be able to hike the popular trail. The chance of being approved to hike Angels Landing is 25%. Find the probability that:
- ▶ in 10 independent drawings, he will be able to have the opportunity to hike at least twice.
 - ▶ he will have a chance to hike within 3 drawings.

A Discrete Uniform

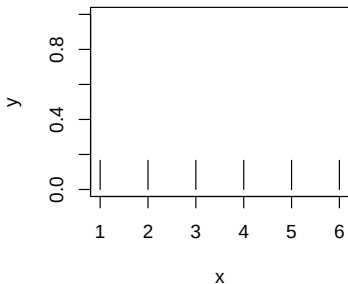
A **discrete uniform** distribution is characterized by having equal chances of each value of the support (e.g. all natural numbers) occurring between a and b , where a is the lower bound of the support and b is the upper bound:

$$P(X = x) = \frac{1}{n}$$

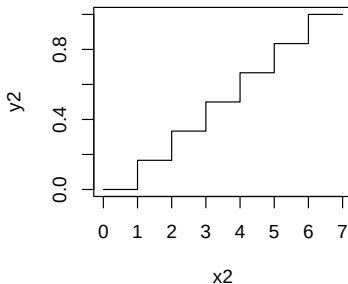
Question: What are some examples of random variables that follow a discrete uniform distribution?

PMF/CDF of Discrete Uniform

$X \sim \text{DUnif}(a = 1, b = 6)$



$X \sim \text{DUnif}(a = 1, b = 6)$



Worksheet

For the rest of the period, work on the worksheet that I'll be passing out.

- Discuss with those around you.
- I'll also be passing out exams at this time.

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